

SAT-MYSQL Test -- 21 Feb 2026.

1) Create Customer Table (Customer id, Name, Mobile no, Address, Bill Amount, Bill date), create a stored procedure to list customers registered within a given date range.?

Create Table.

```
CREATE TABLE Customer (  
  Customer_id INT PRIMARY KEY AUTO-INCREMENT,  
  name VARCHAR(100), mobile_no VARCHAR(15),  
  address VARCHAR(255), bill-amount DECIMAL(10,2);  
  bill-date DATE );
```

Stored Procedure:

```
DELIMITER //
```

```
CREATE PROCEDURE GetCustomersByDate Range (  
  IN start-date DATE,  
  IN end-date DATE.)
```

```
BEGIN
```

```
  SELECT * FROM Customer
```

```
  WHERE bill-date BETWEEN start-date AND end-date;
```

```
END //
```

```
DELIMITER ;
```

CALL Procedure

CALL GetCustomersByDateRange ('2026-01-01', '2026-01-31');

2) Stored Procedure - AVG, MIN, MAX, COUNT, SUM.

DELIMITER //

CREATE PROCEDURE GetBillStatistics (

OUT avg_bill DECIMAL (10, 2),

OUT min_bill DECIMAL (10, 2),

OUT max_bill DECIMAL (10, 2),

OUT total_customer INT,

OUT total_bill DECIMAL (10, 2))

BEGIN

SELECT

AVG (bill-amount),

MIN (bill-amount),

MAX (bill-amount),

COUNT (bill-amount),

SUM (bill-amount)

INTO
avg_bill, min_bill, max_bill, total_customer, total_bill

FROM customer;

END //

DELIMITER ;

Call Procedure.

CALL GetBillStatistics (@avg, @min, @max, @count, @sum);

SELECT @avg, @min, @max, @count, @sum;

3) Find 3rd Highest salary.

```
SELECT DISTINCT salary
```

```
FROM Employee e1
```

```
WHERE 3 = ( SELECT COUNT( DISTINCT salary)
```

```
FROM Employee e2
```

```
WHERE e2.salary >= e1.salary );
```

4) SQL Transaction:

```
START TRANSACTION;
```

```
UPDATE Account
```

```
SET balance = balance - 5000
```

```
where account_id = 1;
```

```
update Account
```

```
SET balance = balance + 5000
```

```
WHERE account_id = 2;
```

```
COMMIT;
```

if error:

```
ROLLBACK.
```

5) Find 4th Highest Bill,

```
select bill-amount
```

```
FROM ( SELECT bill-amount;
```

```
DENSE_RANK () OVER (ORDER BY bill-amount DESC)
```

```
AS rnk FROM customer )
```

```
AS temp.
```

```
where rnk = 4;
```


6) What is view?

* A view is a virtual table created from a SELECT query.

* It does not store data physically

Example.

```
CREATE VIEW HighBillCustomers AS  
SELECT name, bill-amount  
FROM customer  
WHERE bill-amount > 10000;
```

```
SELECT * FROM HighBillCustomers;
```

7) CTE - Employees Earning More than Average Salary,

```
WITH AvgSalary AS (  
    SELECT AVG (salary) AS avg_sal FROM Employee  
)  
SELECT e.*  
FROM Employee e, AvgSalary a  
where e.salary > a.avg_sal;
```

8) Windows Function:

* A window Function Performs calculations across a set of rows related to the current row.

Common Functions:

* RANK()

* DENSE_RANK()

* ROW-NUMBER()

* SUM () OVER()

Example :-

```
SELECT name, salary,  
       RANK () OVER (ORDER BY salary DESC) AS Rank - Position  
FROM Employees.
```

9) After update trigger :-

```
DELIMITER //
```

```
CREATE TRIGGER prevent negative salary  
AFTER UPDATE ON Employee  
FOR EACH ROW  
BEGIN  
    IF NEW-SALARY < 0 THEN  
        SIGNAL SQLSTATE '45000'  
        SET message-text = "salary cannot be  
                                negative";  
    END IF;  
END //
```

```
DELIMITER ;
```

10) what is INDEX :-

- * An index improves the speed of data retrieval in a table
- * It works like a book index

Example :-

```
CREATE INDEX idx-employee-salary  
ON Employee (salary);  
SELECT * FROM Employees where salary = 500000;
```